

Dyadic-Comb Sums for the Fabius–Rvachev System

Rational sample generating functions, exact Euler–Maclaurin collapse, complex powers, and iterated summation

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Abstract

Let F be the Fabius distribution function, let $u = \text{up}$ be Rvachev’s even probability density on $[-1, 1]$, and put $M = 2^N$, $h = M^{-1}$. This paper studies the unweighted dyadic-comb sums

$$L_{N,\alpha} = h \sum_{k=0}^{M-1} \left(\frac{k}{M} \right)^\alpha F\left(\frac{k}{M} \right),$$

where the zero term is omitted for nonintegral or negative α . These are ordinary left Riemann sums, not the self-weighted up-function quadrature already present in the repository corpus.

For every nonnegative integer p , an exact finite Bernoulli–Thue–Morse formula is obtained by inserting the dyadic evaluator for $F(k/2^N)$ and performing the remaining power sums by Faulhaber’s formula. A stronger compression follows from a Bell-polynomial formula for complete signed Thue–Morse power blocks: although the original expression contains 2^N signs, only $p + 2$ Bell terms survive. The entire sample row also has a single rational ordinary generating function. Its apparent pole of order $N + 1$ is cancelled by the finite Thue–Morse product, leaving one simple pole whose constant tail is exactly 1.

The central theorem is an exact Euler–Maclaurin collapse. If $I_p = \int_0^1 x^p F(x) \, dx$, then

$$L_{N,p} = I_p - \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} p^{2r-1} h^{2r}$$

for every $p \leq 2\lfloor N/2 \rfloor$. Thus a normally asymptotic Euler–Maclaurin series becomes a finite exact identity on a degree range that grows with the dyadic level. The proof uses periodic trapezoidal aliasing and the order $N + \nu_2(\ell)$ zeros of the Fabius characteristic function at frequencies $2^N \ell$. At odd levels the first failure is an explicit half-integer sinc-product Dirichlet series whose signs are Thue–Morse signs.

For arbitrary $\alpha \in \mathbb{C}$, the finite comb is entire in α and admits an exact Hurwitz-zeta formula; negative integer powers are therefore exact rational harmonic sums. The corresponding Rvachev combs exhibit a critical logarithmic divergence at $\alpha = -1$ and zeta-dominated power divergences below it. Finally, an n -fold prefix sum is reduced exactly to finitely many shifted one-fold combs through unsigned Stirling numbers of the first kind, and a fractional-order extension is given by the binomial/Gamma summation kernel. Exact rational and high-precision experiments accompany the proofs. Novelty claims are relative to the pinned repository snapshot and a targeted literature search; no unconditional claim of worldwide priority is made.

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1 Scope, corpus audit, and contribution boundary

1.1 Pinned source boundary

The requested source boundary was the recursive LaTeX tree under `Analysis/FabiusFunction/docs`. The audit was pinned to commit

fa53516af8cbe7d792a97b25378b228b96e37cf9,

dated 28 August 2026. The repository’s own source ledger records a stable 67-document core together with live additions, while the pinned tree also contains auxiliary TeX fragments and source-audit copies. The audit therefore proceeded by mathematical lineage rather than by treating byte-identical or embedded copies as independent prior art. In particular, the following consolidated families were used as the principal interfaces:

- the primary Fabius–Rvachev exposition and mathematical glossary;
- the dyadic, q -binomial, asymptotic, and Thue–Morse archive studies;
- the consolidated inverse-and-sampling volume;
- the consolidated integration-and-transform volume;
- both representation-frontier volumes, including their embedded source audits;
- the imported or corrected classical papers and the Lean walkthrough.

A targeted phrase and formula search was then made for ordinary dyadic Riemann sums, unweighted Fabius combs, Euler–Maclaurin exactness, complete row generating functions, and iterated comb sums. No counterpart of the combined theorem package below was found. The external search likewise located classical Fabius arithmetic and compact-support foundations [1, 2], general Euler–Maclaurin theory including singular variants [6, 7], and the standard Bernoulli–Hurwitz special-function machinery [5], but not the same construction.

Meaning of the status labels

Corpus-known means explicitly present in the pinned repository and used here as input. **Proved here** means derived in this report from those inputs and standard analysis. **Experimentally checked** means verified by exact rational or high-precision code but not promoted beyond the accompanying proof. **Conjectural** means that a plausible nonvanishing, sign, or asymptotic assertion remains open.

1.2 Strong negative controls: what is not claimed as new

The repository already contains, in substantially greater generality than needed here, the results summarized in its primary and consolidated volumes [8, 9, 10, 11]:

- exact dyadic values of F through moments, Thue–Morse sums, and Gaussian q -binomial coefficients;
- moment recurrences, Bernoulli cumulants, complete Bell-polynomial formulas, and inverse-dyadic arithmetic;
- the Fourier product for u , integer zero multiplicity $1 + \nu_2(m)$, local zero jets, half-integer signs, and sharp Fourier-decay analyses;
- positive self-weighted shifted quadrature $2^{-N} \sum p((k + \theta)/2^N) u((k + \theta)/2^N)$, exact for $\deg p \leq N$;
- continuous integrals $\int_0^1 x^\alpha F(x) \, dx$, their Mellin continuation, negative powers, Volterra ladders, and fractional integrals;

- repeated integration of Thue–Morse signs, geometric Richardson filters, and inverse-Fabius sampling identities.

The object studied here is different: the lattice is uniform and the summand itself is $x^\alpha F(x)$ or $x^\alpha u(x)$; no sample of u is inserted as an integration weight. The exact Euler–Maclaurin collapse in [section 6](#) is therefore not a restatement of the corpus’s self-sampling quadrature theorem, although [section 8](#) shows that the two theories meet in a useful consistency identity.

1.3 Contribution map

Topic	Result	Status
Sample-row algebra	Rational OGF for the complete level- N Fabius row, with exact pole cancellation and constant tail.	Proved here
Integer powers	Finite Bernoulli–Thue–Morse formula for every N, p , plus a $p + 2$ -term Bell–Prouhet compression.	Proved here
Riemann-sum exactness	Euler–Maclaurin is exact for $p \leq 2\lfloor N/2 \rfloor$.	Proved here
First failure	Odd-level leading defect as a half-integer sinc/Thue–Morse Dirichlet series.	Proved here
Complex powers	Entire finite comb and exact Hurwitz-zeta representation; rational negative-integer specializations.	Proved here
Rvachev combs	Meromorphic continuous limit, critical logarithm at -1 , and zeta-dominated lower powers.	Proved here
Iterated sums	Exact Stirling reduction of an n -fold sum to shifted one-fold combs; Gamma-kernel fractional order.	Proved here
Exact precision	Nonvanishing of the first degree outside the proved range for every level.	Conjectural

Table 1: Main outputs and proof status.

2 Normalization and inherited exact data

2.1 The Fabius law and Rvachev density

Let Y be the random variable

$$Y = \sum_{j=1}^{\infty} 2^{-j} U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1], \quad (2.1)$$

and let

$$F(x) = \mathbb{P}(Y \leq x), \quad 0 \leq x \leq 1.$$

Then

$$F(1 - x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (2.2)$$

and F is flat at both endpoints. Put $X = 2Y - 1$. Its density is Rvachev’s even up-function $u = \text{up}$, normalized by

$$\int_{-1}^1 u(x) \, dx = 1, \quad u(x) = F(1 - |x|), \quad u(x) = 1 - F(x) \quad (0 \leq x \leq 1). \quad (2.3)$$

The Fourier convention used throughout is

$$\Phi(\xi) = \int_{-1}^1 u(x) e^{-2\pi i x \xi} \, dx = \prod_{j=0}^{\infty} \text{sinc}_{\pi} \left(\frac{\xi}{2^j} \right), \quad \text{sinc}_{\pi} z = \frac{\sin(\pi z)}{\pi z}. \quad (2.4)$$

At every nonzero integer m ,

$$\text{ord}_{\xi=m} \Phi = 1 + \nu_2(m). \quad (2.5)$$

The characteristic function of Y is therefore

$$\chi(\xi) = \mathbb{E}e^{-2\pi i \xi Y} = e^{-\pi i \xi} \Phi(\xi/2). \quad (2.6)$$

2.2 Moments and continuous monomial integrals

Write

$$c_r = \int_{-1}^1 x^{2r} u(x) \, dx, \quad c_0 = 1. \quad (2.7)$$

The exact recurrence is

$$(2r+1)(2^{2r}-1)c_r = \sum_{j=0}^{r-1} \binom{2r+1}{2j} c_j. \quad (2.8)$$

By $Y = (X+1)/2$, the complete Fabius moments are

$$d_m := \mathbb{E}Y^m = 2^{-m} \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j. \quad (2.9)$$

For $p \in \mathbb{N}_0$, integration by parts gives

$$I_p := \int_0^1 x^p F(x) \, dx = \frac{1-d_{p+1}}{p+1} = \frac{1}{p+1} \left(1 - 2^{-p-1} \sum_{j=0}^{\lfloor (p+1)/2 \rfloor} \binom{p+1}{2j} c_j \right). \quad (2.10)$$

More generally, if

$$d(s) = \mathbb{E}Y^s, \quad (2.11)$$

then endpoint flatness makes $d(s)$ entire, and

$$I(\alpha) := \int_0^1 x^\alpha F(x) \, dx = \frac{1-d(\alpha+1)}{\alpha+1} \quad (2.12)$$

is entire after removing the apparent singularity at $\alpha = -1$.

2.3 The dyadic value interface

Let

$$M = 2^N, \quad h = M^{-1}, \quad \varepsilon_a = (-1)^{s_2(a)}, \quad (2.13)$$

and define

$$P_N(y) = \sum_{j=0}^{\lfloor N/2 \rfloor} \binom{N}{2j} c_j y^{N-2j}, \quad C_N = \frac{2^{-N(N+1)/2}}{N!}. \quad (2.14)$$

The corpus's exact dyadic evaluator can be written compactly as

$$\boxed{F\left(\frac{k}{M}\right) = C_N \sum_{a=0}^{k-1} \varepsilon_a P_N(2(k-a)-1), \quad 0 \leq k \leq M.} \quad (2.15)$$

The empty sum at $k = 0$ is zero. This finite convolution is the sole arithmetic input needed for [sections 4](#) and [5](#).

3 The dyadic combs

3.1 Left, right, and trapezoidal forms

For $\alpha \in \mathbb{C}$, define the punctured left Fabius comb by

$$L_{N,\alpha} = \frac{1}{M} \sum_{k=1}^{M-1} \left(\frac{k}{M}\right)^\alpha F\left(\frac{k}{M}\right), \quad (3.1)$$

where positive real powers use the real logarithm. For $p \in \mathbb{N}_0$, this agrees with the ordinary left Riemann sum including $k = 0$, because $F(0) = 0$. At fixed N , $L_{N,\alpha}$ is an exponential polynomial and hence an entire function of α .

The right and trapezoidal sums for $x^p F(x)$ are

$$R_{N,p} = L_{N,p} + h, \quad T_{N,p} = L_{N,p} + \frac{h}{2}, \quad (3.2)$$

because $F(0) = 0$ and $F(1) = 1$. The trapezoidal normalization is the natural one for Fourier aliasing.

3.2 Shifted phases

For a phase $\theta \in \mathbb{R}$, one may similarly define

$$L_{N,\alpha}^{(\theta)} = h \sum_{\substack{k \in \mathbb{Z} \\ 0 < k + \theta < M}} \left(\frac{k + \theta}{M}\right)^\alpha F\left(\frac{k + \theta}{M}\right). \quad (3.3)$$

The present report emphasizes the centered phase $\theta = 0$, where all samples are level- N dyadic values. A dyadic phase $\theta = a/2^s$ reduces to a residue class inside the finer level $N + s$; an exact roots-of-unity filter is given in [theorem 4.3](#).

4 A rational generating function for a complete sample row

4.1 Euler operator form

Let

$$\mathcal{D} = z \frac{d}{dz}, \quad \Theta_N(z) = \prod_{j=0}^{N-1} (1 - z^{2^j}) = \sum_{a=0}^{M-1} \varepsilon_a z^a. \quad (4.1)$$

The kernel in [\(2.15\)](#) has OGF

$$\sum_{r \geq 1} P_N(2r - 1) z^r = P_N(2\mathcal{D} - 1) \frac{z}{1 - z}. \quad (4.2)$$

Convolution with the Thue–Morse block therefore gives the following result.

Theorem 4.1 (Complete sample-row generating function). *Define*

$$\mathcal{F}_N(z) := C_N \Theta_N(z) P_N(2\mathcal{D} - 1) \frac{z}{1 - z}. \quad (4.3)$$

Then

$$\mathcal{F}_N(z) = \sum_{k \geq 0} \tilde{F}_N(k) z^k, \quad \tilde{F}_N(k) = \begin{cases} F(k/M), & 0 \leq k \leq M, \\ 1, & k \geq M. \end{cases} \quad (4.4)$$

In particular,

$$A_N(z) := \sum_{k=0}^{M-1} F(k/M) z^k = \mathcal{F}_N(z) - \frac{z^M}{1-z} \quad (4.5)$$

is a polynomial of degree at most $M - 1$.

Proof. The coefficient identity through $k = M$ is exactly (2.15). Applying $(2\mathcal{D} - 1)^r$ to $z/(1 - z)$ produces a rational function with denominator $(1 - z)^{r+1}$ and numerator degree at most $r + 1$. Hence the largest denominator in $P_N(2\mathcal{D} - 1)z/(1 - z)$ is $(1 - z)^{N+1}$. On the other hand,

$$\Theta_N(z) = (1 - z)^N \prod_{j=0}^{N-1} (1 + z + \cdots + z^{2^j-1}). \quad (4.6)$$

Thus $\mathcal{F}_N$ has at most one simple pole at $z = 1$, and its numerator has degree at most

$$(M - 1) + (N + 1) - N = M.$$

Consequently its coefficients are constant from index M onward. The coefficient at index M is $F(1) = 1$, so the constant tail is 1. Subtracting $z^M/(1 - z)$ removes it and leaves A_N . $\square$

Pole collapse

The finite Thue–Morse product does more than enforce Prouhet cancellation. It removes *exactly* N of the $N + 1$ poles generated by the degree- N polynomial kernel. The surviving simple pole encodes the saturation $F(k/M) = 1$ after the right endpoint.

Corollary 4.2 (Natural powers by Euler differentiation). *For every $p \in \mathbb{N}_0$,*

$$L_{N,p} = M^{-p-1} (\mathcal{D}^p A_N)(z)|_{z=1}. \quad (4.7)$$

If the spectral Euler power is defined on polynomials by

$$\mathcal{D}^\alpha \left(\sum_{k \geq 0} a_k z^k \right) := \sum_{k \geq 1} k^\alpha a_k z^k, \quad (4.8)$$

then the same formula holds for every $\alpha \in \mathbb{C}$:

$$L_{N,\alpha} = M^{-\alpha-1} (\mathcal{D}^\alpha A_N)(1). \quad (4.9)$$

Corollary 4.3 (Dyadic phase as a roots-of-unity filter). *Let $\theta = a/2^s$, put $D = N + s$, and let $\omega = \exp(2\pi i/2^s)$. Then*

$$L_{N,\alpha}^{(a/2^s)} = \frac{2^{-D\alpha}}{M2^s} \sum_{r=0}^{2^s-1} \omega^{-ar} (\mathcal{D}^\alpha A_D)(\omega^r), \quad (4.10)$$

with the obvious endpoint convention.

Proof. The usual character projector $2^{-s} \sum_r \omega^{r(k-a)}$ selects indices $k \equiv a \pmod{2^s}$ from the finer level- D row. $\square$

4.2 Low-level examples

The exact rational functions begin

$$\mathcal{F}_1(z) = \frac{z(z+1)}{2(1-z)}, \quad (4.11)$$

$$\mathcal{F}_2(z) = \frac{z(z+1)(z+5)(5z+1)}{72(1-z)}, \quad (4.12)$$

$$\mathcal{F}_3(z) = \frac{z(z+1)^3(z^2+1)(z^2+16z+1)}{288(1-z)}. \quad (4.13)$$

The numerator is reciprocal after removal of the initial z , reflecting $F(k/M) + F((M-k)/M) = 1$. A systematic factorization theory for these row numerators is proposed in [section 13](#).

5 Closed forms for nonnegative integer powers

5.1 Bernoulli–Thue–Morse formula

Insert (2.15) into (3.1), exchange the finite sums, and put $r = k - a$. Expanding $(a+r)^p$ and $(2r-1)^d$ leaves ordinary power sums $\sum_{r=1}^{M-1-a} r^q$, which are Bernoulli-polynomial differences.

Theorem 5.1 (Exact closed form for every N, p). *For $p \in \mathbb{N}_0$,*

$$L_{N,p} = C_N M^{-p-1} \sum_{a=0}^{M-1} \varepsilon_a H_{N,p}(a), \quad (5.1)$$

where

$$H_{N,p}(a) = \sum_{j=0}^{\lfloor N/2 \rfloor} \binom{N}{2j} c_j \sum_{r=0}^p \binom{p}{r} a^{p-r} \sum_{s=0}^{N-2j} \binom{N-2j}{s} 2^s (-1)^{N-2j-s} \times \frac{B_{r+s+1}(M-a) - B_{r+s+1}(1)}{r+s+1}. \quad (5.2)$$

Here $B_n(x)$ denotes the Bernoulli polynomial. The term $a = M-1$ vanishes automatically.

Proof. From (2.15),

$$M^{p+1} C_N^{-1} L_{N,p} = \sum_{a=0}^{M-2} \varepsilon_a \sum_{r=1}^{M-1-a} (a+r)^p P_N(2r-1).$$

Use

$$(a+r)^p = \sum_{q=0}^p \binom{p}{q} a^{p-q} r^q, \quad (2r-1)^d = \sum_{s=0}^d \binom{d}{s} 2^s (-1)^{d-s} r^s,$$

and Faulhaber's identity

$$\sum_{r=1}^K r^m = \frac{B_{m+1}(K+1) - B_{m+1}(1)}{m+1}. \quad (5.3)$$

Taking $K = M-1-a$ gives (5.2). □

Remark 5.2. Formula (5.1) is manifestly rational. It is useful for exact evaluation at moderate N , but it still contains a complete 2^N -term signed block. The next subsection removes that exponential-looking part.

5.2 Bell–Prouhet collapse

Define the signed block moments

$$T_{N,m} = \sum_{a=0}^{M-1} \varepsilon_a a^m. \quad (5.4)$$

Their exponential generating function is

$$\sum_{m \geq 0} T_{N,m} \frac{t^m}{m!} = \Theta_N(e^t) = \prod_{j=0}^{N-1} (1 - e^{2^j t}). \quad (5.5)$$

Factor each term as

$$1 - e^x = -x \exp(x/2) \frac{\sinh(x/2)}{x/2}.$$

This gives a closed logarithmic jet.

Theorem 5.3 (Bell formula for complete Thue–Morse power blocks). *Let*

$$\lambda_{N,1} = \frac{M-1}{2}, \quad (5.6)$$

$$\lambda_{N,2r} = \frac{B_{2r}}{2r} \frac{2^{2rN} - 1}{2^{2r} - 1} \quad (r \geq 1), \quad (5.7)$$

$$\lambda_{N,2r+1} = 0 \quad (r \geq 1). \quad (5.8)$$

Then

$$\Theta_N(e^t) = (-1)^N 2^{N(N-1)/2} t^N \exp \left(\sum_{r \geq 1} \lambda_{N,r} \frac{t^r}{r!} \right). \quad (5.9)$$

Consequently,

$$T_{N,m} = 0 \quad (m < N), \quad (5.10)$$

and, for $s \geq 0$,

$$T_{N,N+s} = (-1)^N 2^{N(N-1)/2} \frac{(N+s)!}{s!} Y_s(\lambda_{N,1}, \dots, \lambda_{N,s}), \quad (5.11)$$

where Y_s is the complete exponential Bell polynomial.

Proof. The standard expansion

$$\log \frac{e^x - 1}{x} = \frac{x}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r} \frac{x^{2r}}{(2r)!} \quad (5.12)$$

may be summed over $x = 2^j t$. The leading factors contribute $(-1)^N 2^{0+1+\dots+(N-1)} t^N$, the linear terms give $(M-1)t/2$, and the even terms give the finite geometric sum in (5.7). Comparing with

$$\exp \left(\sum_{r \geq 1} x_r \frac{t^r}{r!} \right) = \sum_{s \geq 0} Y_s(x_1, \dots, x_s) \frac{t^s}{s!}$$

proves the coefficient formula. □

Corollary 5.4 (Only $p + 2$ signed moments survive). *The polynomial $H_{N,p}(a)$ has degree at most $N + p + 1$. Therefore*

$$L_{N,p} = C_N M^{-p-1} \sum_{s=0}^{p+1} [a^{N+s}] H_{N,p}(a) T_{N,N+s}, \quad (5.13)$$

with $T_{N,N+s}$ given by (5.11). Thus the original full Thue–Morse block collapses to at most $p + 2$ Bell-polynomial terms, independently of N .

Proof. In (5.2), the Bernoulli polynomial has degree $r + s + 1$; after multiplication by a^{p-r} , the degree is at most $p + s + 1 \leq p + N + 1$. All monomials below degree N vanish by (5.10). $\square$

A useful computational dichotomy

For fixed N and many degrees p , the row polynomial A_N is convenient. For fixed p and large symbolic N , the Bell collapse is preferable: its signed part has constant length $p + 2$, and all dependence on the exponentially long block is compressed into the explicit jets $\lambda_{N,r}$.

6 Exact Euler–Maclaurin collapse

6.1 Periodic trapezoidal aliasing

Put

$$g_p(x) = x^p F(x), \quad 0 \leq x \leq 1, \quad (6.1)$$

and let

$$\widehat{g}_p(n) = \int_0^1 g_p(x) e^{-2\pi i n x} dx. \quad (6.2)$$

The periodic extension of g_p has endpoint jump 1. Its Fourier series takes the midpoint value at the jump, so the trapezoidal alias identity is

$$T_{N,p} = \sum_{\ell \in \mathbb{Z}} \widehat{g}_p(M\ell). \quad (6.3)$$

The zero-frequency term is I_p .

A probability-layer computation makes every nonzero Fourier coefficient explicit.

Lemma 6.1 (Fourier coefficient through the Fabius characteristic function). *For $n \in \mathbb{Z} \setminus \{0\}$,*

$$\widehat{g}_p(n) = \sum_{j=0}^p \frac{p^j}{(2\pi i n)^{j+1}} \left(\frac{\chi^{(p-j)}(n)}{(-2\pi i)^{p-j}} - 1 \right). \quad (6.4)$$

Proof. Since $F(x) = \mathbb{P}(Y \leq x)$, Fubini gives

$$\widehat{g}_p(n) = \mathbb{E} \int_Y^1 x^p e^{-2\pi i n x} dx. \quad (6.5)$$

Repeated integration by parts, using $e^{-2\pi i n} = 1$, yields

$$\int_Y^1 x^p e^{-2\pi i n x} dx = \sum_{j=0}^p \frac{p^j}{(2\pi i n)^{j+1}} (Y^{p-j} e^{-2\pi i n Y} - 1).$$

Finally, $\mathbb{E}[Y^r e^{-2\pi i n Y}] = \chi^{(r)}(n)/(-2\pi i)^r$. $\square$

6.2 The exact remainder

The -1 terms in (6.4) can be summed by Euler’s formula for $\zeta(2r)$; they are precisely the ordinary Euler–Maclaurin boundary corrections. The remaining characteristic-function terms form an exact spectral remainder.

Theorem 6.2 (Exact Euler–Maclaurin plus an alias remainder). *For every $N \geq 0$ and $p \in \mathbb{N}_0$,*

$$L_{N,p} = I_p - \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} p^{2r-1} h^{2r} + \mathcal{R}_{N,p}, \quad (6.6)$$

where

$$\mathcal{R}_{N,p} = \sum_{\ell \in \mathbb{Z} \setminus \{0\}} \sum_{j=0}^p \frac{p^j}{(2\pi i M \ell)^{j+1}} \frac{\chi^{(p-j)}(M\ell)}{(-2\pi i)^{p-j}}. \quad (6.7)$$

The series is absolutely convergent after pairing and, equivalently, is the Fourier series remainder of the periodic trapezoidal rule.

Proof. Insert (6.4) into (6.3). The characteristic terms give (6.7). For the boundary terms, only odd $j = 2r - 1$ survive the symmetric sum over ℓ , and

$$-\frac{p^{2r-1}}{(2\pi i M)^{2r}} \sum_{\ell \neq 0} \ell^{-2r} = \frac{B_{2r}}{(2r)!} p^{2r-1} M^{-2r}.$$

Finally $T_{N,p} = L_{N,p} + h/2$. □

6.3 Growing exactness range

At the dual lattice, (2.6) and (2.5) give, for $N \geq 1$,

$$\text{ord}_{\xi=M\ell} \chi = N + \nu_2(\ell), \quad \ell \neq 0. \quad (6.8)$$

This forces almost every term in (6.7) to vanish.

Theorem 6.3 (Exact Euler–Maclaurin collapse). *Let*

$$d_N = 2 \left\lfloor \frac{N}{2} \right\rfloor. \quad (6.9)$$

Then, for every $p \in \mathbb{N}_0$ with $p \leq d_N$,

$$L_{N,p} = I_p - \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} p^{2r-1} h^{2r}. \quad (6.10)$$

Equivalently, the level- N dyadic comb agrees exactly with its complete finite Euler–Maclaurin polynomial through degree N when N is even, and through degree $N - 1$ when N is odd.

Proof. The case $N = 0, p = 0$ follows from symmetry: $L_{0,0} = 0 = I_0 - 1/2$. Suppose $N \geq 1$. If $p < N$, every derivative order $p - j$ occurring in (6.7) is below the zero order $N + \nu_2(\ell)$, so $\mathcal{R}_{N,p} = 0$.

It remains to consider $p = N$. Aliases with even ℓ still vanish. At odd ℓ , only the $j = 0$ term can survive. Since Φ is even,

$$\chi^{(N)}(-M\ell) = (-1)^N \chi^{(N)}(M\ell).$$

The denominator in the $j = 0$ term changes sign under $\ell \mapsto -\ell$. Hence positive and negative odd aliases cancel exactly when N is even. For odd N , no such cancellation is forced. This is precisely the range $p \leq 2\lfloor N/2 \rfloor$. □

Remark 6.4 (Why this is not ordinary polynomial exactness). The summand contains the nonpolynomial function F . The theorem does not say that a fixed quadrature rule integrates all polynomials against a fixed weight. Instead, the boundary polynomial generated by Euler–Maclaurin becomes exact because the nonzero Fourier aliases are killed by increasingly high-order zeros of χ . It is a spectral cancellation theorem for a family of ordinary Riemann sums.

Corollary 6.5 (Low-degree formulas). *With $h = 2^{-N}$, the formulas listed in [table 2](#) are exact once the displayed level threshold is met.*

p	I_p	sufficient N	exact $L_{N,p}$
0	$\frac{1}{2}$	0	$\frac{1}{2} - \frac{h}{2}$
1	$\frac{13}{36}$	2	$\frac{13}{36} - \frac{h}{2} + \frac{1}{12}h^2$
2	$\frac{5}{18}$	2	$\frac{5}{18} - \frac{h}{2} + \frac{1}{6}h^2$
3	$\frac{1207}{5400}$	4	$\frac{1207}{5400} - \frac{h}{2} + \frac{1}{4}h^2 - \frac{1}{120}h^4$
4	$\frac{251}{1350}$	4	$\frac{251}{1350} - \frac{h}{2} + \frac{1}{3}h^2 - \frac{1}{30}h^4$
5	$\frac{22661}{142884}$	6	$\frac{22661}{142884} - \frac{h}{2} + \frac{5}{12}h^2 - \frac{1}{12}h^4 + \frac{1}{252}h^6$
6	$\frac{16427}{119070}$	6	$\frac{16427}{119070} - \frac{h}{2} + \frac{1}{2}h^2 - \frac{1}{6}h^4 + \frac{1}{42}h^6$
7	$\frac{63446897}{520506000}$	8	$\frac{63446897}{520506000} - \frac{h}{2} + \frac{7}{12}h^2 - \frac{7}{24}h^4 + \frac{1}{12}h^6 - \frac{1}{240}h^8$
8	$\frac{7096441}{65063250}$	8	$\frac{7096441}{65063250} - \frac{h}{2} + \frac{2}{3}h^2 - \frac{7}{15}h^4 + \frac{2}{9}h^6 - \frac{1}{30}h^8$

Table 2: Exact Euler–Maclaurin formulas generated and checked by the companion script.

6.4 The first odd-level defect

At odd N , the first degree outside the exact range is $p = N$, and only odd aliases contribute. The local factorization of Φ at $m = 2^{N-1}q$, q odd, gives

$$\Phi^{(N)}(2^{N-1}q) = -\frac{N!}{(2^{N-1}q)^N} \Phi(q/2). \quad (6.11)$$

Theorem 6.6 (First odd-level defect). *If $N \geq 1$ is odd, then*

$$\mathcal{R}_{N,N} = \frac{2(-1)^{(N+1)/2}N!}{(2\pi)^{N+1}2^{N(N+1)}} \sum_{m=0}^{\infty} \frac{\Phi((2m+1)/2)}{(2m+1)^{N+1}}. \quad (6.12)$$

Moreover,

$$\operatorname{sgn} \Phi((2m+1)/2) = (-1)^{s_2(m)} = \varepsilon_m, \quad (6.13)$$

so the defect is a Thue–Morse-signed half-integer sinc-product Dirichlet series.

Proof. For $p = N$, retain only $j = 0$ and odd ℓ in (6.7). Since $M = 2^N$, differentiation of (2.6) at the order- N zero gives

$$\chi^{(N)}(M\ell) = 2^{-N} \Phi^{(N)}(2^{N-1}\ell) = -N!2^{-N^2}\ell^{-N} \Phi(\ell/2).$$

Substitution and pairing ℓ with $-\ell$ give (6.12). The sign law (6.13) follows by counting the integer zeros crossed by the real even product Φ . $\square$

N	first degree beyond the proved range	exact defect
1	1	$-1/144$
2	3	$-97/1382400$
3	3	$23/22118400$
4	5	$9671/11985978654720$
5	5	$-1543/767102633902080$
6	7	$-9903527/146509414227756711936000$
7	7	$1196113/37506410042305718255616000$
8	9	$21035141003/590082134533477495427314445451264000$

Table 3: Exact first defects through level eight. The table supports the exact-precision conjecture in Conjecture 13.1.

$N \setminus p$	0	1	2	3	4	5	6	7	8	9	10
0	✓	.	.	.	.	.	.	.	.	.	.
1	✓	.	.	.	.	.	.	.	.	.	.
2	✓	✓	✓	.	.	.	.	.	.	.	.
3	✓	✓	✓	.	.	.	.	.	.	.	.
4	✓	✓	✓	✓	✓	.	.	.	.	.	.
5	✓	✓	✓	✓	✓	.	.	.	.	.	.
6	✓	✓	✓	✓	✓	✓	✓	.	.	.	.
7	✓	✓	✓	✓	✓	✓	✓	.	.	.	.
8	✓	✓	✓	✓	✓	✓	✓	✓	✓	.	.

Table 4: Exact rational check of (6.10). A check mark means that the complete Euler–Maclaurin polynomial equals the direct dyadic sum exactly.

7 Complex, fractional, and negative powers

7.1 Exact Hurwitz-zeta form

The Bernoulli formula is the natural specialization of a more general finite Hurwitz-zeta identity. For $s \in \mathbb{C}$,

$$\sum_{k=a+1}^{M-1} k^s = \zeta(-s, a+1) - \zeta(-s, M). \quad (7.1)$$

The pole of each individual zeta function cancels in the difference, which is an entire function of s .

Theorem 7.1 (Exact complex-power comb). *For every $\alpha \in \mathbb{C}$,*

$$L_{N,\alpha} = C_N M^{-\alpha-1} \sum_{a=0}^{M-2} \varepsilon_a \sum_{j=0}^{\lfloor N/2 \rfloor} \binom{N}{2j} c_j \sum_{r=0}^{N-2j} \binom{N-2j}{r} 2^r (-1)^{N-2j-r} \\ \times \sum_{q=0}^r \binom{r}{q} (-a)^{r-q} [\zeta(-\alpha-q, a+1) - \zeta(-\alpha-q, M)]. \quad (7.2)$$

The right side is entire in α , and at $\alpha = p \in \mathbb{N}_0$ it reduces to [theorem 5.1](#) by $\zeta(-m, x) = -B_{m+1}(x)/(m+1)$.

Proof. After exchanging the k - and a -sums as in the proof of [theorem 5.1](#), expand

$$(2(k-a)-1)^d = \sum_{r=0}^d \binom{d}{r} 2^r (-1)^{d-r} \sum_{q=0}^r \binom{r}{q} k^q (-a)^{r-q}.$$

The remaining finite sum of $k^{\alpha+q}$ is [\(7.1\)](#). □

Corollary 7.2 (Negative integer powers are rational). *For $q \in \mathbb{N}$,*

$$L_{N,-q} = M^{q-1} \sum_{k=1}^{M-1} \frac{F(k/M)}{k^q} \in \mathbb{Q}. \quad (7.3)$$

The case $q = 1$ is the harmonic Fabius comb $\sum_{k=1}^{M-1} F(k/M)/k$.

7.2 Euler–Maclaurin for fixed complex exponent

For fixed α , the function $x^\alpha F(x)$ extends smoothly and flatly through $x = 0$, because F and all of its derivatives decay faster than every power there. At $x = 1$, all positive derivatives of F vanish, so

$$\left. \frac{d^m}{dx^m} (x^\alpha F(x)) \right|_{x=1} = \alpha^m. \quad (7.4)$$

Standard Euler–Maclaurin therefore gives an all-orders expansion, locally uniformly in α .

Theorem 7.3 (Complex-power asymptotic). *For α in any compact subset of $\mathbb{C}$,*

$$L_{N,\alpha} \sim I(\alpha) - \frac{h}{2} + \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} \alpha^{2r-1} h^{2r}, \quad N \rightarrow \infty, \quad (7.5)$$

with a remainder after R terms that is $\mathcal{O}_K(h^{2R+2})$ on every compact $K \subset \mathbb{C}$. Equivalently, for every prescribed algebraic order A , subtracting sufficiently many displayed terms leaves $\mathcal{O}_K(h^A)$.

Remark 7.4. When $\alpha = p \in \mathbb{N}_0$, the formal series terminates. The unusual content of [theorem 6.3](#) is that the remaining alias remainder is not merely small: it is identically zero once N crosses a degree-dependent threshold.

For $\alpha = -1$,

$$L_{N,-1} \sim I(-1) - \frac{h}{2} - \sum_{r \geq 1} \frac{B_{2r}}{2r} h^{2r}. \quad (7.6)$$

The table below illustrates convergence for several nonintegral exponents.

α	$L_{4,\alpha}$	$L_{6,\alpha}$	$L_{8,\alpha}$	EM estimate of $I(\alpha)$
-1	0.726663799849	0.750407155874	0.756285603841	0.758240000404
-0.5	0.575973485504	0.599563638374	0.605432549961	0.607386310743
0.5	0.389880928801	0.413165845835	0.419015684123	0.42096817334
1.5	0.283877847065	0.306857575817	0.312688340557	0.314639558208
3.5	0.17301875663	0.195388418551	0.201181037431	0.203129711955

Table 5: Direct complex-power combs and a six-term reversed Euler–Maclaurin estimate. The last column is numerical evidence, not an independent closed-form evaluation.

7.3 A Newton-series alternative

Let

$$A_N(z) = \sum_{k=1}^{M-1} F(k/M) z^k. \quad (7.7)$$

Because k^α has the Newton expansion

$$k^\alpha = \sum_{r=0}^k \binom{k}{r} \Delta^r 0^\alpha, \quad (7.8)$$

one may write

$$L_{N,\alpha} = M^{-\alpha-1} \sum_{r=0}^{M-1} \Delta^r 0^\alpha \sum_{k=r}^{M-1} \binom{k}{r} F(k/M). \quad (7.9)$$

This is finite at every N and gives a second entire interpolation in α . It is potentially advantageous near negative integers, where the Hurwitz-zeta representation contains large cancelling terms.

8 Rvachev combs and the critical exponent

8.1 Punctured absolute comb

On $[0, 1]$, $u(x) = 1 - F(x)$. Define the pure power comb

$$P_{N,\alpha} = h \sum_{k=1}^{M-1} (k/M)^\alpha = M^{-\alpha-1} [\zeta(-\alpha) - \zeta(-\alpha, M)]. \quad (8.1)$$

The punctured symmetric absolute Rvachev comb is

$$U_{N,\alpha}^\times := h \sum_{0 < |k| < M} |k/M|^\alpha u(k/M). \quad (8.2)$$

Proposition 8.1 (Fabius–Rvachev comb relation). *For every $\alpha \in \mathbb{C}$,*

$$\boxed{U_{N,\alpha}^\times = 2(P_{N,\alpha} - L_{N,\alpha})}. \quad (8.3)$$

If the principal branch is used for x^α on the negative axis, then the signed complex monomial comb is

$$h \sum_{0 < |k| < M} (k/M)^\alpha u(k/M) = (1 + e^{\pi i \alpha}) (P_{N,\alpha} - L_{N,\alpha}). \quad (8.4)$$

Corollary 8.2 (Recovery of exact self-sampling moments). *For $r \geq 1$ and $2r \leq N$,*

$$U_{N,2r}^\times = c_r. \quad (8.5)$$

For $r = 0$, adding the central sample $hu(0) = h$ gives

$$h + U_{N,0}^\times = 1. \quad (8.6)$$

Proof. This is the centered phase of the corpus-known self-sampling quadrature for u . It also follows directly by subtracting the exact Faulhaber sum from (6.10); the Bernoulli boundary polynomials cancel. $\square$

Consistency bridge

The corpus's self-weighted quadrature and the present unweighted Fabius comb are distinct constructions, but $u = 1 - F$ makes them complementary. Exact up-moments are equivalent to cancellation between an ordinary Faulhaber sum and the exact Euler–Maclaurin polynomial for $L_{N,2r}$.

8.2 Continuous Rvachev Mellin transform

For $\Re \alpha > -1$,

$$J(\alpha) := \int_0^1 x^\alpha u(x) \, dx = \frac{1}{\alpha + 1} - I(\alpha). \quad (8.7)$$

Since $u(x) = 1 + \mathcal{O}(x^A)$ for every $A > 0$ as $x \downarrow 0$, J extends meromorphically to $\mathbb{C}$ with exactly one pole:

$$\text{Res}_{\alpha=-1} J(\alpha) = 1. \quad (8.8)$$

The full absolute moment is $2J(\alpha)$.

8.3 Critical and supercritical negative powers

At $\alpha = -1$, (8.3) becomes

$$U_{N,-1}^\times = 2(H_{M-1} - L_{N,-1}), \quad (8.9)$$

where H_n is the harmonic number. The Euler–Maclaurin expansions of H_{M-1} and $L_{N,-1}$ have identical algebraic tails, hence cancel term by term.

Theorem 8.3 (Critical logarithm). *As $N \rightarrow \infty$,*

$$U_{N,-1}^\times = 2 \log M + 2(\gamma - I(-1)) + \mathcal{O}(M^{-A}) \quad (8.10)$$

for every fixed $A > 0$. Thus the only divergence is the universal logarithm generated by $u(0) = 1$; the Fabius correction contributes the finite constant $-2I(-1)$.

Proof. Use

$$H_{M-1} = \log M + \gamma - \frac{h}{2} - \sum_{r=1}^R \frac{B_{2r}}{2r} h^{2r} + \mathcal{O}(h^{2R+2})$$

and compare with (7.6). Since R is arbitrary, all algebraic terms cancel. $\square$

For an integer $q > 1$, generalized harmonic asymptotics similarly give

$$U_{N,-q}^\times = 2\zeta(q)M^{q-1} - 2\left(\frac{1}{q-1} + I(-q)\right) + \mathcal{O}(M^{-A}) \quad (8.11)$$

for every fixed $A > 0$. Again all algebraic Euler–Maclaurin corrections beyond the finite-part constant cancel. This is a discrete Mellin-renormalization phenomenon: $2\zeta(q)M^{q-1}$ is the lattice singularity, while $-2((q-1)^{-1} + I(-q))$ is the finite part of the Rvachev integral.

9 Iterated sums of arbitrary integer order

9.1 Definition and binomial kernel

For a sequence $a = (a_k)_{k \geq 0}$, let

$$(\Sigma a)_m = \sum_{k=0}^m a_k, \quad \Sigma^n = \underbrace{\Sigma \circ \cdots \circ \Sigma}_{n \text{ times}}. \quad (9.1)$$

Take

$$a_k^{(N,\alpha)} = (k/M)^\alpha F(k/M), \quad 0 \leq k < M, \quad (9.2)$$

with the punctured convention at $k = 0$. Define the normalized n -fold comb by

$$L_{N,\alpha}^{[n]} := h^n (\Sigma^n a^{(N,\alpha)})_{M-1}. \quad (9.3)$$

Repeated prefix summation gives

$$L_{N,\alpha}^{[n]} = h^n \sum_{k=1}^{M-1} \binom{M-k+n-2}{n-1} (k/M)^\alpha F(k/M). \quad (9.4)$$

Equivalently, if $A_{N,\alpha}(z) = \sum_{k=1}^{M-1} (k/M)^\alpha F(k/M) z^k$, then

$$(\Sigma^n a)_m = [z^m] \frac{A_{N,\alpha}(z)}{(1-z)^n}. \quad (9.5)$$

For $\alpha = p \in \mathbb{N}_0$, one may take $A_{N,p}(z) = M^{-p} \mathcal{D}^p A_N(z)$.

9.2 Reduction to shifted one-fold combs

Use

$$\binom{M-k+n-2}{n-1} = \frac{1}{(n-1)!} (M-k)^{\overline{n-1}} = \frac{1}{(n-1)!} \sum_{j=0}^{n-1} \begin{bmatrix} n-1 \\ j \end{bmatrix} (M-k)^j, \quad (9.6)$$

where $\begin{bmatrix} r \\ j \end{bmatrix}$ is an unsigned Stirling number of the first kind.

Theorem 9.1 (Exact n -fold reduction). *For $n \geq 1$ and every $\alpha \in \mathbb{C}$,*

$$L_{N,\alpha}^{[n]} = \frac{M^{1-n}}{(n-1)!} \sum_{j=0}^{n-1} \begin{bmatrix} n-1 \\ j \end{bmatrix} M^j \sum_{\ell=0}^j (-1)^\ell \binom{j}{\ell} L_{N,\alpha+\ell}. \quad (9.7)$$

Thus an n -fold sum requires only the n shifted base combs $L_{N,\alpha}, \dots, L_{N,\alpha+n-1}$.

Proof. Expand $(M-k)^j$ in (9.6). Since

$$\sum_{k=1}^{M-1} k^\ell (k/M)^\alpha F(k/M) = M^{\ell+1} L_{N,\alpha+\ell}, \quad (9.8)$$

substitution into (9.4) gives the stated powers of M . □

Corollary 9.2 (Closed form for natural powers). *For $p \in \mathbb{N}_0$, substitute either [theorem 5.1](#) or [theorem 6.2](#) into (9.7). In particular, if*

$$N \geq 2 \left\lceil \frac{p+n-1}{2} \right\rceil, \quad (9.9)$$

then every shifted comb in (9.7) is given by the exact finite Euler–Maclaurin polynomial (6.10). Hence $L_{N,p}^{[n]}$ is an explicit rational polynomial in h , with coefficients formed from $I_p, \dots, I_{p+n-1}$, Bernoulli numbers, and Stirling numbers.

9.3 Continuous limit

From (9.4),

$$h^{n-1} \binom{M-k+n-2}{n-1} \rightarrow \frac{(1-x)^{n-1}}{(n-1)!}, \quad x = k/M. \quad (9.10)$$

Therefore

$$\lim_{N \rightarrow \infty} L_{N,\alpha}^{[n]} = \frac{1}{(n-1)!} \int_0^1 (1-x)^{n-1} x^\alpha F(x) \, dx \quad (9.11)$$

for every $\alpha \in \mathbb{C}$, and binomial expansion gives

$$\lim_{N \rightarrow \infty} L_{N,\alpha}^{[n]} = \frac{1}{(n-1)!} \sum_{\ell=0}^{n-1} (-1)^\ell \binom{n-1}{\ell} I(\alpha + \ell). \quad (9.12)$$

This is the ordinary Volterra kernel of order n evaluated at the right endpoint.

10 Fractional-order summation

The generating-function identity $(1-z)^{-n}$ has a canonical positive-real continuation. For $\rho \in \mathbb{C}$ with $\Re \rho > 0$, define

$$L_{N,\alpha}^{[\rho]} := h^\rho \sum_{k=1}^{M-1} \frac{\Gamma(M-k+\rho-1)}{\Gamma(\rho)\Gamma(M-k)} (k/M)^\alpha F(k/M). \quad (10.1)$$

The coefficient is $[z^{M-1-k}](1-z)^{-\rho}$, so

$$L_{N,\alpha}^{[\rho]} = h^\rho [z^{M-1}] (1-z)^{-\rho} A_{N,\alpha}(z). \quad (10.2)$$

At $\rho = n \in \mathbb{N}$, this is exactly (9.4).

Proposition 10.1 (Discrete semigroup). *Let Σ^ρ denote convolution with the coefficients of $(1-z)^{-\rho}$. Wherever the finite sums are defined,*

$$\Sigma^\rho \Sigma^\sigma = \Sigma^{\rho+\sigma}. \quad (10.3)$$

Proof. Multiply generating functions: $(1-z)^{-\rho}(1-z)^{-\sigma} = (1-z)^{-\rho-\sigma}$. $\square$

Theorem 10.2 (Fractional Volterra limit). *For $\Re \rho > 0$ and α in compact subsets of $\mathbb{C}$,*

$$\lim_{N \rightarrow \infty} L_{N,\alpha}^{[\rho]} = \frac{1}{\Gamma(\rho)} \int_0^1 (1-x)^{\rho-1} x^\alpha F(x) \, dx. \quad (10.4)$$

Proof idea. Uniform Stirling asymptotics for the Gamma ratio give

$$h^{\rho-1} \frac{\Gamma(M-k+\rho-1)}{\Gamma(\rho)\Gamma(M-k)} \rightarrow \frac{(1-x)^{\rho-1}}{\Gamma(\rho)}.$$

Endpoint flatness at $x = 0$, together with $\Re \rho > 0$ at $x = 1$, supplies a uniform integrable majorant after splitting off finitely many boundary nodes. $\square$

Remark 10.3. For nonintegral ρ , there is no finite Stirling reduction analogous to (9.7). One promising replacement is a convergent Newton series in shifted combs $L_{N,\alpha+j}$, with coefficients obtained from the generalized binomial kernel. This is developed as an open direction in [section 13](#).

11 Arithmetic consequences

The corpus supplies explicit common denominator bounds for all level- N dyadic values. One convenient nonsharp bound is

$$\Delta_N = N! 2^{N(N+1)/2} (2 \lfloor N/2 \rfloor + 1)!! \prod_{j=1}^{\lfloor N/2 \rfloor} (2^{2j} - 1), \quad (11.1)$$

for which $\Delta_N F(k/M) \in \mathbb{Z}$ at every level- N node.

Proposition 11.1 (Immediate denominator bounds). *For $p \in \mathbb{N}_0$,*

$$\Delta_N M^{p+1} L_{N,p} \in \mathbb{Z}. \quad (11.2)$$

For $q \in \mathbb{N}$, if $\Lambda_M = \text{lcm}(1, \dots, M-1)$, then

$$\Delta_N \Lambda_M^q L_{N,-q} \in \mathbb{Z}. \quad (11.3)$$

More generally, the same bounds propagate through (9.7), with the additional factorial and powers of M visible there.

Remark 11.2. These bounds are deliberately crude. The exact Euler–Maclaurin formula often produces large cancellations between the moment denominator in I_p and the Bernoulli powers of 2. The sharp 2-adic and odd-denominator valuations appear to be governed by an interaction among Mersenne factors $2^{2j} - 1$, Bernoulli denominators, and the Prouhet block. See [Problem 13.4](#).

12 Numerical and symbolic verification

The companion program `dyadic_comb_experiments.py` uses three independent exact routes:

1. direct dyadic samples from (2.15);
2. the exchanged Bernoulli–Thue–Morse formula (5.1);
3. the Bell–Prouhet coefficient collapse (5.13).

It also constructs the rational OGF (4.3) symbolically, checks the exact Euler–Maclaurin range through level eight, compares direct and reduced iterated sums, and evaluates nonintegral exponents at high precision.

Reproducibility

Running

```
python dyadic_comb_experiments.py --outdir generated --max-level 8
```

performs all self-checks and regenerates the CSV and LaTeX tables included in this paper. All exact checks use `Fractions.Fraction`; SymPy is used for Bernoulli polynomials, Bell coefficient extraction, and rational-function simplification; mpmath is used only for nonintegral powers.

The exact checks performed by the script include

$$L_{N,p}^{\text{direct}} = L_{N,p}^{\text{Bernoulli}} = L_{N,p}^{\text{Bell}} \quad (12.1)$$

for representative ranges, and

$$L_{N,p}^{[n],\text{direct}} = L_{N,p}^{[n],\text{Stirling}} \quad (12.2)$$

for $N \leq 6$, $p \leq 4$, and $n \leq 4$. No floating-point comparison is used to assert an exact theorem.

13 Conjectures and frontier directions

13.1 Exact degree of Euler–Maclaurin collapse

The proved range is $p \leq d_N = 2\lfloor N/2 \rfloor$. Exact computations through $N = 8$ show a nonzero defect at the first possible degree beyond it.

Conjecture 13.1 (Exact precision). *For every $N \geq 1$,*

$$\mathcal{R}_{N,p} = 0 \quad (0 \leq p \leq d_N), \quad \mathcal{R}_{N,d_N+1} \neq 0. \quad (13.1)$$

Equivalently, the Euler–Maclaurin collapse has exact maximal degree $2\lfloor N/2 \rfloor$.

For odd N , this reduces to nonvanishing of the explicit series (6.12). The first term is positive in magnitude and the Fourier product decays rapidly, suggesting a first-harmonic domination argument. A rigorous proof must control possible Thue–Morse-signed cancellation uniformly in N . For even N , the first defect mixes the first two odd-alias jets with the leading twice-odd alias layer; the local Bell-polynomial zero-jet calculus from the corpus should make this finite-layer structure explicit.

13.2 Sign and asymptotic size of the first defect

Conjecture 13.2 (Odd-level sign). *For odd N , the sum in (6.12) is positive. Hence*

$$\text{sgn } \mathcal{R}_{N,N} = (-1)^{(N+1)/2}. \quad (13.2)$$

A stronger target is an asymptotic expansion dominated by $\Phi(1/2)$:

$$\mathcal{R}_{N,N} \sim \frac{2(-1)^{(N+1)/2} N! \Phi(1/2)}{(2\pi)^{N+1} 2^{N(N+1)}}. \quad (13.3)$$

The correction is exponentially small in N from the odd denominators and is further suppressed by the sinc-product decay. The repository’s sharp arithmetic-ray estimates should be adaptable to a quantitative proof.

13.3 Row-polynomial geometry

Write

$$\mathcal{F}_N(z) = \frac{Q_N(z)}{1-z}, \quad \deg Q_N \leq M. \quad (13.4)$$

The first examples exhibit reciprocal factors, powers of $1+z$, and unexpectedly small-coefficient palindromic pieces. Natural problems are:

Problem 13.3. Determine the exact reciprocal symmetry of Q_N , the multiplicities of cyclotomic factors, and whether the noncyclotomic zeros obey a limiting curve or annulus law.

A route through $F(k/M) + F((M - k)/M) = 1$ gives a functional equation for $A_N(z)$, while (4.3) expresses Q_N through the Thue–Morse mask and the moment polynomial $P_N(2\mathcal{D} - 1)$. The collision of those two descriptions may lead to a finite spectral factorization.

13.4 Sharp denominator valuations

Problem 13.4. Determine

$$\nu_2(\text{den } L_{N,p}) \quad \text{and} \quad v_\ell(\text{den } L_{N,p}) \quad (13.5)$$

for odd primes ℓ , uniformly in N, p . Do the same for the iterated sums and negative powers.

The exact Euler–Maclaurin formula reduces the stable range to the arithmetic of I_p , Bernoulli denominators, and powers of two. Outside that range, the Bell–Prouhet formula adds only $p + 2$ signed moments and may expose a tractable valuation filtration.

13.5 Double scaling near negative powers

For fixed q , section 8 gives the complete leading behavior. A different regime arises when $q = q_N$ grows with N . The smallest lattice node contributes

$$M^{q_N-1} F(1/M), \quad (13.6)$$

and $F(2^{-N})$ has a log-squared/Lambert- W_{-1} asymptotic. Balancing that term against the next nodes should produce a phase diagram in the ratio q_N/N , with transitions governed by a discrete saddle over the lattice index. This is a natural meeting point of the repository’s endpoint asymptotics and the present comb calculus.

13.6 Fractional summation as a two-variable special function

The object

$$(\rho, \alpha) \longmapsto L_{N,\alpha}^{[\rho]} \quad (13.7)$$

is entire in α and meromorphic in ρ before cancellation. Promising directions include:

- a Newton series in shifted $L_{N,\alpha+j}$ for nonintegral ρ ;
- analytic continuation in ρ after subtracting the first lattice jets;
- a Mellin–Barnes representation for the Gamma kernel;
- a uniform double limit $N \rightarrow \infty, \rho = \rho_N$, including large-order fractional Volterra asymptotics;
- exact arithmetic at rational ρ , where Gamma values may organize into controlled transcendental factors times rational dyadic data.

13.7 Base- b and digital generalizations

The Bell–Prouhet mechanism depends only on a finite digital product and a polynomial kernel. Replace $\prod_{j < N} (1 - z^{2^j})$ by a base- b digit mask or a root-of-unity weighted product. Whenever the mask has a zero of order N at $z = 1$, the same argument yields a short Bell collapse. The corresponding probability law would be a geometrically weighted sum of uniform or discrete digits. One may ask which masks also create high-order zeros at the dual lattice, and hence an exact Euler–Maclaurin range.

13.8 Formalization roadmap

The most modular Lean targets are finite and algebraic:

1. formalize the convolution OGF (4.3) and the pole-cancellation degree bound;
2. formalize (5.9) as a finite-product identity and derive (5.11);
3. formalize the Bernoulli sum exchange (5.1);
4. build the periodic trapezoidal alias theorem for a function with one endpoint jump;
5. combine it with the already formalized integer zero orders of Φ to prove theorem 6.3;
6. formalize the finite Stirling reduction (9.7);
7. only then add the analytic Hurwitz and fractional-order layers.

The exactness theorem is particularly suitable for formalization because its essential spectral argument is finite after the zero-order lemma: every forbidden alias derivative is literally zero.

14 Conclusion

The ordinary dyadic Riemann sums of $x^p F(x)$ hide considerably more structure than a generic discretization error. Their exact dyadic arithmetic produces a rational sample-row generating function; Prouhet cancellation compresses exponentially many signs to finitely many Bell polynomials; and the same dyadic geometry forces an exact Euler–Maclaurin formula on a degree range that grows linearly with the level. The first failure is not an opaque remainder but a controlled sinc-product Dirichlet series, and negative or fractional powers remain accessible through a finite Hurwitz-zeta formula. Iterated summation adds no essentially new arithmetic obstruction: unsigned Stirling numbers reduce it to a short list of shifted base combs.

The strongest immediate next step is to prove exact maximality of the degree range. At odd levels this is a concrete positivity/noncancellation problem for the half-integer Fourier product; at even levels it is a finite competition among the first two 2-adic alias layers. Both are tightly connected to machinery already developed in the repository, making them realistic frontier targets rather than vague extrapolations.

A Derivation of the row symmetry

Let

$$A_N(z) = \sum_{k=0}^{M-1} F(k/M) z^k. \quad (\text{A.1})$$

Using $F((M-k)/M) = 1 - F(k/M)$, one obtains

$$\begin{aligned} z^M A_N(z^{-1}) &= \sum_{k=1}^M F((M-k)/M) z^k \\ &= \sum_{k=1}^M (1 - F(k/M)) z^k \\ &= \frac{z(1 - z^M)}{1 - z} - A_N(z) - z^M. \end{aligned} \quad (\text{A.2})$$

Equivalently,

$$A_N(z) + z^M A_N(z^{-1}) = \frac{z(1 - z^{M-1})}{1 - z}. \quad (\text{A.3})$$

This exact reciprocal relation explains the palindromic features visible in the low-level row numerators.

B First few moments and continuous integrals

The recurrence (2.8) begins

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}, \quad c_4 = \frac{132809}{32531625}. \quad (\text{B.1})$$

From (2.9),

$$d_1 = \frac{1}{2}, \quad d_2 = \frac{5}{18}, \quad d_3 = \frac{1}{6}, \quad d_4 = \frac{143}{1350}, \quad d_5 = \frac{19}{270}. \quad (\text{B.2})$$

Consequently (2.10) yields the constants displayed in table 2.

C Algorithmic summary

For exact computation of $L_{N,p}$:

1. compute $c_0, \dots, c_{\lfloor N/2 \rfloor}$ from (2.8);
2. choose one of three routes:
 - (a) evaluate all dyadic samples by (2.15) and sum directly;
 - (b) evaluate (5.1);
 - (c) form $H_{N,p}(a)$, retain only coefficients of degrees N through $N + p + 1$, and apply (5.11);
3. when $p \leq 2\lfloor N/2 \rfloor$, bypass all dyadic samples and use the exact Euler–Maclaurin polynomial (6.10);
4. for an n -fold sum, first compute $L_{N,p}, \dots, L_{N,p+n-1}$, then apply (9.7).

The fourth route is asymptotically constant-time in N at fixed p , apart from the cost of the moments entering I_p .

D Source and literature notes

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